

Van Phan

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EDUCATION

Georgia State University

Atlanta, GA

B.S. Mathematics, GPA: 4.12/4.30

Expected Graduation: May, 2027

- *Relevant Coursework:* Linear Algebra, Differential Equations, Mathematical Biology, Biostatistics, Optimization, Numerical Analysis, Analysis, Modern Algebra, Probability and Statistics, Applied Combinatorics, Calculus, Computer Science

RESEARCH PROJECTS

Density and BI-RADS - Aware Triage and Automatic Report Generation for Mammography

- Developed a system to automatically generate clinical reports and prioritize high-risk mammography cases, optimizing clinical workflows and reducing time-to-diagnosis for vulnerable patients by directing attention to high-risk density/severity categories.
- Achieved high-fidelity automated report generation as measured by a **ROUGE-L score of 0.6693**, a **METEOR score of 0.7187**, and a **Word-Level F1 score of 0.6789**, by fine-tuning the **Google MedGemma-1.5-4B Vision-Language Model**, using **Low-Rank Adaptation (LoRA)** and **4-bit quantization** from 510 mammography images and clinical reports.
- Accelerated clinical review process as measured by the shift of **mean rank of high-risk cases from 215.3 to 61.1** in total of 500 cases, by applying Density and BI-RADS Aware Triage method, reducing delays in high-volume screening environments.

Modeling Wound Healing Without Migration and Proliferation

- Established a mathematical framework for Age-related Macular Degeneration (AMD) therapeutics, addressing a leading cause of vision loss by modeling Retinal Pigment Epithelium (RPE) wound healing.
- Simulated purse-string contraction and cell fusion repair mechanisms in Python-based CompuCell3D, by utilizing a **Q-Potts cellular model** and **Monte Carlo evaluation** of the **Hamiltonian free energy states**.
- Proved that **higher cellular volume constraints** yield **statistically faster wound closure**, quantifying intervention viability by analyzing texture tensors and spatial points across different algorithmic iterations.

Enhanced Physics-Informed Neural Networks for Collective Cancer Invasion

- Engineered a **Physics-Informed Neural Networks (PINNs)** model to simulate collective cancer invasion, resulting in a data-efficient, mesh-free solution for complex **Partial Differential Equations (PDEs)**.
- Developed a framework utilizing **Tensorized Fourier Neural Operators (TFNO)** and **PyTorch's autograd engine** to encode physical laws (PDEs, boundary, and initial conditions) directly into the model's loss function, reducing reliance on labeled data.
- Explored advanced techniques, including a **sequence-to-sequence (Seq2seq) PINO** approach and **Augmented Lagrangian methods (ALM)**, to overcome model failure modes like vanishing gradients and improve solution accuracy and stability.

RESEARCH PAPER

- [PrePrint] **DB-ATRG: Density and BI-RADS - Aware Triage and Automatic Report Generation System for Mammography**, published on [arXiv](#).

POSTERS & PRESENTATIONS

Density and BI-RADS - Aware Triage and Automatic Report Generation System for Mammography,

- Poster presented at Georgia State Undergraduate Research Conference, 2026.

Enhanced Physics-Informed Neural Networks for Collective Cancer Invasion,

- Best project award at Summer Undergraduate Research Symposium, 2025.

Morphological Feature Analysis of Retinal Pigment Epithelial Cell from Mice during Aging,

- Poster presented at Georgia State Undergraduate Research Conference, 2025.

Modeling Wound Healing Without Migration and Proliferation,

- Poster presented at Summer Undergraduate Research Symposium, 2024.

RESEARCH AND WORK EXPERIENCE

RIMMES Program

Sep. 2025 – Present

Undergraduate Researcher (Advisor: Dr. Russell Jeter)

Atlanta, GA

- Led a 3-member student team in DB-ATRG project, developing a system to automatically report clinical report and prioritize high-risk cases in mammography.

Georgia State University

Aug. 2025 – Present

Chemistry Laboratory Teaching Assistant

Atlanta, GA

- Mentored 50+ undergraduate students in Survey of Chemistry I and II, improving understanding of fundamental principles and reducing procedural errors through one-on-one guidance and demonstration of proper laboratory techniques.

Center for the Advancement of Students and Alumni

Jun. 2024 – August. 2025

Undergraduate Researcher (Advisor: Dr. Yi Jiang)

Atlanta, GA

- Led a 3-member student team in two concurrent computational biology projects, developing wound healing simulations and morphological analysis of Retinal Pigment Epithelial (RPE) cells.

Near-peer Mentor

- Provided near-peer mentorship and research support to students in the Math Path Program, guiding research technique implementation and project development for the Enhanced Physics-Informed Neural Networks for Collective Cancer Invasion project.

SELECTED PROJECTS

Morphological Feature Analysis of Retinal Pigment Epithelial Cell from Mice during Aging

- Developed an automated end-to-end machine learning pipeline for classifying Retinal Pigment Epithelial (RPE) cell during aging, generating insights for potential therapeutic strategies in Age-related Macular Degeneration.
- Achieved highly accurate classification as measured by **90%+ cross-validation F1-score**, by extracting **133 morphological and texture-based features** from 326 RPE cell images and implementing a **stacking ensemble** of **XGBoost, LightGBM, and CatBoost** with a Logistic Regression meta-learner.
- Identified critical RPE aging biomarkers, as demonstrated by top feature importance rankings, suggesting **Local Binary Patterns (LBP)** and **geometric shape metrics** (circularity, aspect ratio) as the most influential classification variables.

HONORS & AWARDS

- **Best Project Award** at Summer Undergraduate Research Symposium (2025)
- **President's List** at Georgia State University (2023, 2024, 2025, 2026)
- **Prospect Award** at Vietnam Science and Engineering Fair (ViSEF) (2022), National Level
- **First Place** at Vietnam Science and Engineering Fair (ViSEF) (2022), City Level
- **Consolation Prize** at Engineering and Science Fair of Ho Chi Minh City University of Technology (2022)
- **Bronze Prize** in Physics at the Ho Chi Minh City Excellent Student Contest (2019)

TECHNICAL SKILLS

- **Programming & Tools:** Python, Julia, R, MATLAB, SQL, Git/GitHub, Linux/CLI, LaTeX, CompuCell3D, Microsoft Office Suite
- **Scientific & Math Modeling:** Physics-Informed Neural Networks, PDE Modeling, Tensorized Fourier Neural Operators, Augmented Lagrangian Method, Q-Potts Cellular Modeling, Monte Carlo Evaluation
- **ML, LLM & AI:** LLMs, Fine-Tuning, Quantization, Transformers, TRL, Hugging Face, Ensemble Methods
- **Libraries:** PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy, SciPy, Scikit-image, Pillow, Mahotas, Matplotlib, Seaborn